SET C - QUESTION 1 (20 marks)

a) Diagram 1 shows a parallelogram PQRS drawn on a Cartesian plane.

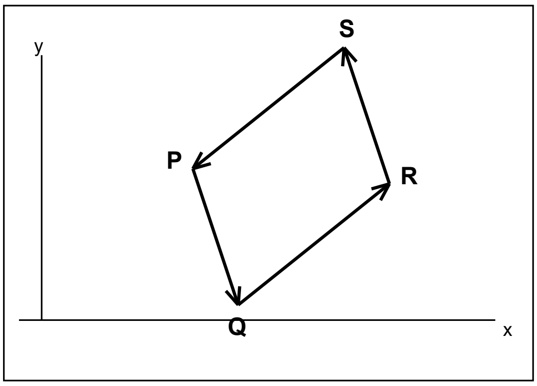

Diagram 1: Parallelogram PQRS

It is given that PQ = 4i + 9j and QR = 7i - 3j. Find PR.

b) The following information refers to the vectors p and q.

p = (12 , 2k-3) q = (4 , 5)

It is given that p = h q, where p is parallel to q and h is a constant. Calculate the value of,

i. h

ii. k

c) i. It is given that RA = 2i + 3j and RB = 10i - 3j. Find,

(a) AB

(b) the unit vector in the direction of AB.

ii. Given XY = 5i + tj, where t is a constant and XY is parallel to ZW = 15i - 18j. Find the value of t.

d) In Diagram 2, ABCD is a quadrilateral, diagonals AC and BD intersect at point O.

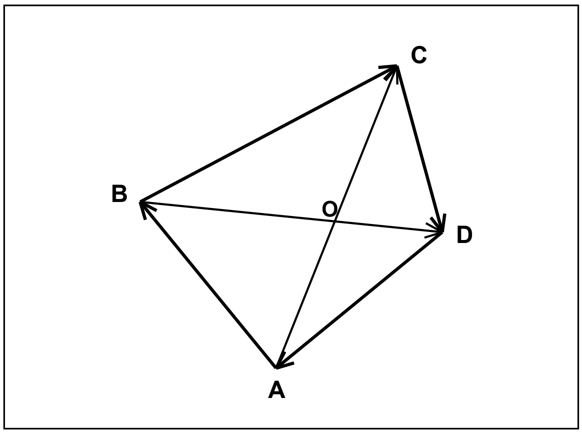

Diagram 2: Quadrilateral ABCD

It is given that BO : OD = 2 : 3, AB = 5p, AD = 5q, and BC = 7p + 8q.

i. Express in terms of p and q

(a) BD

(b) AO

ii. Find the ratio for AO : OC. [4 marks]

[Total: 20 marks]

SET D - QUESTION 1 (20 marks)

a) Diagram 1 shows a parallelogram TUVW drawn on a Cartesian plane.

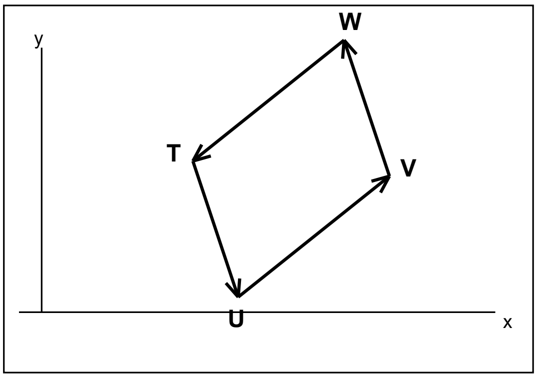

Diagram 1: Parallelogram TUVW

It is given that TU = 6i + 5j and UV = 2i - 8j. Find TV.

b) The following information refers to the vectors p and q.

p = (8 , 3k+2)   q = (2 , 5)

It is given that p = h q, where p is parallel to q and h is a constant. Calculate the value of,

i. h

ii. k

c) i. It is given that RA = 3i + j and RB = 10i + 25j. Find,

(a) AB

(b) the unit vector in the direction of AB.

ii. Given XY = 9i + tj, where t is a constant and XY is parallel to ZW = 27i - 15j. Find the value of t.

d) In Diagram 2, EFGH is a quadrilateral, diagonals EG and FH intersect at point O.

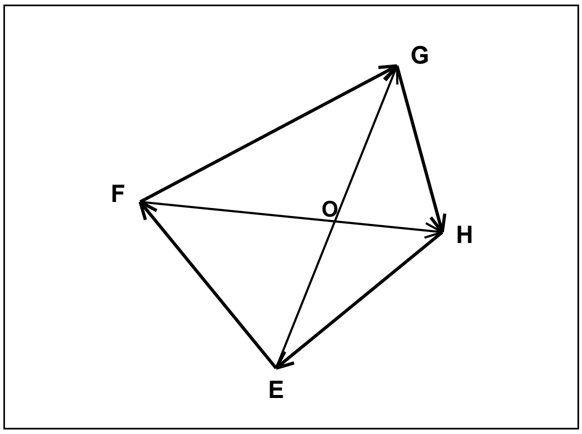

Diagram 2: Quadrilateral EFGH

It is given that FO : OH = 1 : 2, EF = 6p, EH = 6q, and FG = 8p + 7q.

i. Express in terms of p and q

(a) FH

(b) EO

ii. Find the ratio for EO : OG. [4 marks]

[Total: 20 marks]

SET E - QUESTION 1 (20 marks)

a) Diagram 1 shows a parallelogram DEFG drawn on a Cartesian plane.

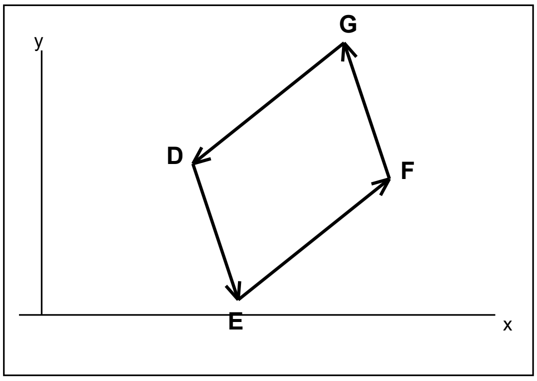

Diagram 1: Parallelogram DEFG

It is given that DE = 3i - 7j and EF = 9i + 2j. Find DF.

b) The following information refers to the vectors p and q.

p = (15 , 2k-1)   q = (3 , 5)

It is given that p = h q, where p is parallel to q and h is a constant. Calculate the value of,

i. h

ii. k

c) i. It is given that RA = 2i - 5j and RB = 22i + 10j. Find,

(a) AB

(b) the unit vector in the direction of AB.

ii. Given XY = 12i + tj, where t is a constant and XY is parallel to ZW = 4i - 10j. Find the value of t.

d) In Diagram 2, JKLM is a quadrilateral, diagonals JL and KM intersect at point O.

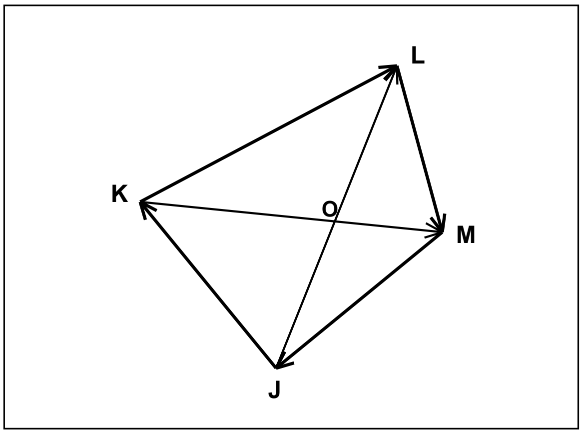

Diagram 2: Quadrilateral JKLM

It is given that KO : OM = 1 : 4, JK = 5p, JM = 10q, and KL = 5p + 5q.

i. Express in terms of p and q

(a) KM

(b) JO

ii. Find the ratio for JO : OL. [4 marks]

[Total: 20 marks]

SET F - QUESTION 1 (20 marks)

a) Diagram 1 shows a parallelogram WXYZ drawn on a Cartesian plane.

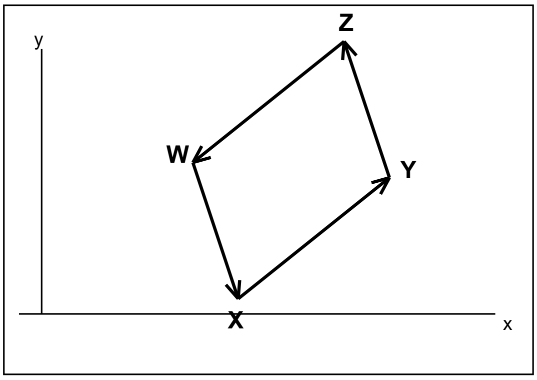

Diagram 1: Parallelogram WXYZ

It is given that WX = 5i + 11j and XY = 6i - 4j. Find WY.

b) The following information refers to the vectors p and q.

p = (6 , 4k-1) q = (2 , 9) (column vectors)

It is given that p = h q, where p is parallel to q and h is a constant. Calculate the value of,

i. h

ii. k

c) i. It is given that RA = 4i + j and RB = 19i + 9j. Find,

(a) AB

(b) the unit vector in the direction of AB.

ii. Given XY = 21i + tj, where t is a constant and XY is parallel to ZW = 7i - 4j. Find the value of t.

d) In Diagram 2, RSTU is a quadrilateral, diagonals RT and SU intersect at point O.

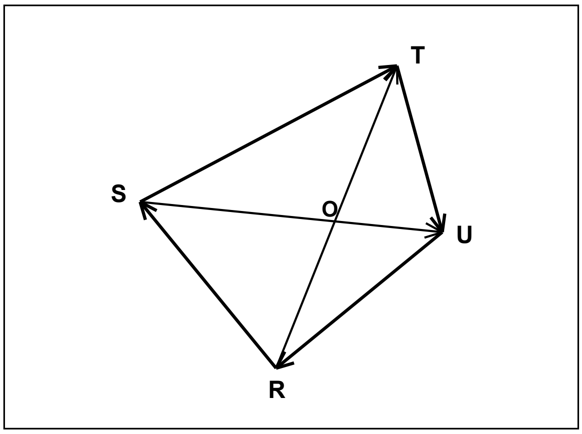

Diagram 2: Quadrilateral RSTU

It is given that SO : OU = 1 : 3, RS = 4p, RU = 8q, and ST = 14p + 12q.

i. Express in terms of p and q

(a) SU

(b) RO

ii. Find the ratio for RO : OT.

[Total: 20 marks]

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